Section 1.1

I. Absolute value function  Definition  $|a| = \begin{cases} a & a \ge 0 \\ -a & a < 0 \end{cases}$

Interpretation: distance

II. Interval Notation

III. Solve Equation / Inequality

Examples

IV. Distance Formula  $d(P_1(x_1, y_1),\ P_2(x_2, y_2)) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Midpoint Formula  $M = \left( \dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2} \right)$

Examples

V. Equation of Circle  Center $C(h, k)$  Radius $r$

Center-Radius Form  $(x - h)^2 + (y - k)^2 = r^2$

General Form  $x^2 + y^2 + Dx + Ey + F = 0$

Examples

## VI. Solve Trigonometric Equations

Examples